

Eco-evo conflicts in a stage-structured mutualism: modeling the consequences of ontogenetic variation in host quality

A.Raine Detmer^{a,*}, Craig W. Osenberg^b, Adrian C. Stier^a, Holly V. Moeller^a

^a Department of Ecology, Evolution, and Marine Biology, University of California, Santa Barbara, California 93106 USA

^b Odum School of Ecology, University of Georgia, 140 E Green St, Athens, GA, 30602, USA

ARTICLE INFO

Keywords:

Mutualism

Partner choice

Eco-evo conflicts

Adaptive dynamics

Coral-fish interactions

ABSTRACT

Stage-structured mutualisms occur when mutualistic interactions vary across the ontogeny of one or both interacting species. They are widespread and can play important roles in ecological communities, yet little is known about the evolution of partner choice in stage-structured mutualisms. Inspired by branching corals and associated damselfish, we modeled a stage-structured mutualism between a large, long-lived host species and a smaller mutualistic partner. We explored the evolution of partner choice behavior (strength of partner's preference for adult hosts) when host quality increased with host size and the partner either increased host survival (a defensive mutualism) or increased host growth (a nutritional mutualism). We found that while partners always evolved strong preferences for adult (high quality) hosts, evolved behaviors often resulted in lower equilibrium population sizes and longer post-disturbance recovery times relative to simulations in which partner preferences were fixed at weaker values. These differences were greatest for nutritional mutualisms, indicating that the type of mutualistic interaction was an important determinant of whether a strong preference for adult hosts had ecological consequences at the population level. Overall, our results suggest ontogenetic variation in host quality can create tension between what is best for individual partners (avoiding low quality hosts) and what is best for partner populations (spreading benefits across host life stages), highlighting the importance of considering both the evolutionary and ecological dynamics of stage-structured mutualisms.

1. Introduction

Mutualisms, interactions in which two or more species benefit from their association with one another, are important and widespread in ecosystems (Bronstein 2015). Mutualistic interactions can confer substantial benefits to the species involved; for example, the symbiosis between stony corals and zooxanthellae algae underpins some of the most productive ecosystems on Earth (Muscattine and Porter 1977). However, the strength of these benefits often varies and may shift toward neutrality or even become detrimental under certain environmental or ecological conditions (Hay et al., 2004; Chamberlain et al. 2014). One major determinant of the outcome of a potentially mutualistic interaction is mutualist quality. There is often a diversity of available mutualists both within and across species that may differ greatly in the strength of the benefits or costs they provide (Batstone et al. 2018). Given that associations with costly species can potentially have large fitness consequences (Bronstein 2001; Zhang et al. 2021; Leal et al. 2025), partner choice, i.e., the selection of a mutualist(s) from available

candidates, has become a major focus of mutualism research (Bull and Rice 1991; Bshary and Grutter 2002; Frederickson 2013; Younginger and Friesen 2019).

Existing research on partner choice generally considers scenarios in which candidate mutualists include multiple species (e.g., fungal species in plant-mycorrhizal associations, strains of nitrogen-fixing bacteria in legume rhizobia systems, fungal strains cultivated by fungus-growing ants; Bever et al., 2009; Simms and Taylor 2002; Mueller et al. 2004). Some of this work also considers within-species partner choice, such as different strains of mutualistic microbes that belong to the same species; however, for mutualisms between multicellular species, there is growing recognition that ontogenetic variation (variation across life stages) is an important and relatively understudied source of within-species variation in mutualist quality (Nakazawa 2020). For instance, juvenile crayfish actively remove cleaner worms whereas adults tolerate them, likely because these worms are primarily parasites of gill tissue in juveniles but provide a net benefit to larger crayfish (Skelton et al. 2014; Thomas et al. 2016). Such ontogenetic variation in the costs and benefits of

* Corresponding author at: 2111 Noble Hall, UCSB, Santa Barbara, CA, USA, 93106.

E-mail address: adetmer@ucsb.edu (A.Raine Detmer).

mutualistic interactions has been widely documented (see Nakazawa 2020 for a full review), raising many questions about the role of stage-specific partner choice in such systems.

Stage-specific partner choice is likely most relevant for mutualisms where interactions persist over multiple life stages of at least one of the species involved. This might include, for example, mutualisms between a larger, longer-lived species (hereafter referred to as the host; e.g., trees, corals) and a smaller, shorter-lived species (hereafter referred to as the partner). Here, we will define “partner choice” as a single partner species choosing a host, rather than a host choosing from among potential partners. Existing evolutionary models of partner choice in these systems predict that when ontogenetic variation in host quality is large, partners usually evolve to associate with either adult or juvenile hosts depending on which stage is most beneficial (Nakazawa et al. 2023). More broadly, studies of partner choice tend to focus on what is best for an individual (i.e., which decisions will maximize an individual partner’s fitness)—with the assumption that a partner should associate with hosts that provide it with the greatest benefits (Bull and Rice 1991; Sachs et al. 2004). However, in the case of stage-structured mutualisms, partners that associate with and thereby benefit a specific host life stage may indirectly affect other host life stages (e.g., by increasing the rate at which juvenile hosts mature into adults; Nakazawa et al. 2023). By influencing the host population’s stage structure, these ecological feedbacks could affect partner and host population dynamics. To our knowledge, the ecological effects of stage-specific partner choice, and whether these depend on the type of mutualistic interaction (e.g., nutritional benefits affecting host growth versus defensive benefits affecting host survival), has yet to be explored in detail.

In this study, we used a theoretical model of a size-structured host and its mutualistic partner to investigate the evolution of stage-specific partner choice and the ecological consequences of these behaviors. We focused on a long-lived host that provides habitat for a smaller partner species, with the assumptions that host quality from the perspective of the partner increases with host size (e.g., larger hosts provide better habitat/shelter) and that the partner benefits the host by either increasing its survival (a defensive mutualism) or its growth rate (a nutritional mutualism). These scenarios are representative of many important mutualisms involving foundation species and species that live on or in them, such as branching corals and associated fishes (Holbrook et al. 2008; Chase et al. 2014; Stier et al. 2025) or acacia trees and ants (Janzen 1966; Orona-Tamayo and Heil 2013). Our model was specifically based on coral-fish mutualisms, both because the mechanisms of these mutualistic interactions are relatively well-studied (e.g., Holbrook et al. 2008, Holbrook and Schmitt 2002) and because of the potential importance of these mutualisms for coral restoration and conservation (Ladd et al. 2018; Stier et al. 2025). We used an adaptive dynamics framework to predict the evolution of partner choice (strength of preference for adult hosts) in this system. To evaluate the population-level effects of evolved partner behaviors, we also calculated the trait values that maximized population-level partner density. These partner-maximizing trait values provided a point of reference that allowed us to determine whether the evolutionary stable behaviors that were best for individual partners were ever “harmful” to the population over ecological timescales (i.e., resulted in lower population sizes than were theoretically possible for a given parameter set). To compare evolved partner behaviors predicted by our model to a real system, we analyzed empirical data on the distribution of coral-associated damselfish across branching corals of varying sizes in Moorea,¹ French Polynesia and investigated whether there was evidence of a preference for

adult corals in this system. Finally, to further explore the ecological consequences of evolved partner preferences, we compared the modeled system’s ability to recover from disturbance under evolutionarily stable versus partner-maximizing behaviors. Overall, this work provides further insight into the evolution of partner choice in stage-structured mutualisms and highlights the potential for conflicts between fitness- and population-maximizing behaviors in these types of interactions.

2. Model description

To explore the ecological and evolutionary dynamics of partner choice in systems where host quality increases across its ontogeny, we modeled populations of a stage-structured host H (e.g., branching coral) and its mutualistic partner P (e.g., coral-associated damselfish) (Fig. 1). In this mutualism, partners either increase their host’s survival (defensive mutualism) or increase their host’s growth (nutritional mutualism), while hosts increase partner survival, with the strength of this effect increasing with host size. This model is based on existing frameworks for modeling stage-specific host-partner systems (Nakazawa et al. 2023; Nakazawa and Katayama 2020), but it expands on them by including more than two host life stages and explicitly considering alternative mechanisms of the mutualistic interaction (specifically, defensive and nutritional mutualisms). The components of our model are described in the following sections.

2.1. Host dynamics in the absence of the partner

To generate a stage-structured host population, we modeled three host stages: juveniles (H_1) that are not yet reproductively mature, small adults (H_2) that are reproductively mature but still growing, and large adults (H_3) that are near the maximum possible host size. Stages are defined based on their size: the first stage ranged from the area of the smallest juveniles A_{min} to the threshold for reproductive maturity A_{rep} , the second stage ranged from A_{rep} to 90 % of the maximum host size A_{max} , and the third stage ranged from 90 % of A_{max} to A_{max} . Our boundaries were roughly based on those reported for *Pocillopora* corals (Stimson 1978; Speare et al., 2022); see Table 1 for default values of all model parameters. We note that our approach equates host stage and size, and our model is therefore most appropriate for hosts in which ontogenetic stages are primarily coupled to body size rather than hosts that undergo dramatic metamorphoses over the course of their ontogeny, such as insects or amphibians.

At each timestep, a proportion S_{Hi} of hosts in stage i survive and, for stages 1 and 2, a fraction G_i of the surviving hosts grow into the next stage. Each individual in stages 2 and 3 produces F_2 and F_3 new juvenile hosts, respectively. Thus, the number of host individuals H_i in each stage i at time $t + 1$ is given by

$$\vec{H}_{t+1} = \begin{bmatrix} (1 - G_1)S_{H1} & F_2 & F_3 \\ G_1S_{H1} & (1 - G_2)S_{H2} & 0 \\ 0 & G_2S_{H2} & S_{H3} \end{bmatrix} \begin{bmatrix} H_1 \\ H_2 \\ H_3 \end{bmatrix}_t \quad (1)$$

We assumed that host survival is reduced by density-independent (e.g., environmental stress) and density-dependent (e.g., competition for resources, disease) mortality and that density-dependent mortality scales with the proportion of available habitat covered by hosts. Thus, survival in each stage can be expressed as a variation of the Ricker model (Ricker 1954):

$$S_{Hi} = \exp\left(-\left(n_{Hi} + m_{Hi}\frac{A_{tot}}{A_{patch}}\right)\right) \quad (2)$$

where n_{Hi} and m_{Hi} are the density-independent and density-dependent mortality rates in stage i , A_{tot} is the total area covered by hosts, and A_{patch} is the area of the entire habitat (i.e., A_{tot}/A_{patch} is the proportion of the habitat covered by the host). We approximated A_{tot} by multiplying the midpoint of each stages’ size range $A_{mid,i}$ by the number of host

¹ With respect to the spelling of Moorea, we followed the *Raapoto* transcription system that is adhered to by a large segment of the Tahitian community, but also recognize other community members follow the Te Fare Vanā’a transcription system where the island name is spelled with an ‘eta (Mo’orea) (see mcr.lternet.edu/spelling_of_Tahitian_place_names).

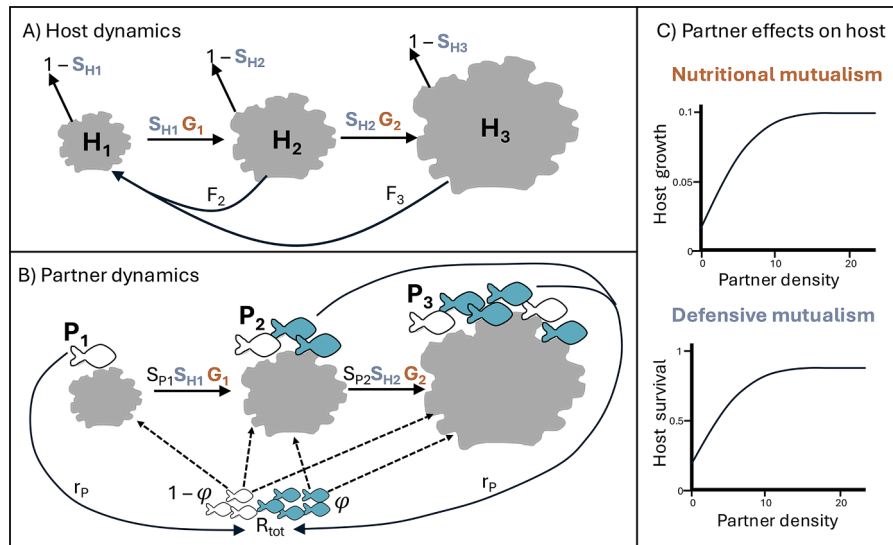


Fig. 1. A-B) Model diagram, with hosts (H) and partners (P) symbolized by coral and damselfish, respectively. S , G , and F represent the survival, growth, and fecundity terms in the transition matrices (for clarity, arrows representing retention within stages are not shown, nor are partner mortality arrows). Partners produce new recruits at a rate r_P , resulting in a total of R_{tot} new recruits that then choose a host stage on which to settle. A fraction ϕ of partner recruits (blue) only settle on adult hosts (H_2 and H_3), while the remaining fraction $1-\phi$ (white) settle on all host stages. C) Partners increase either host growth (nutritional mutualism) or host survival (defensive mutualism). Note that example values are shown on the axes to provide context for our model, but these values would depend on the system of interest and the strength of the partner's effects.

individuals in that stage, H_i , and summing these products across all stages:

$$A_{tot} = A_{mid,1}H_1 + A_{mid,2}H_2 + A_{mid,3}H_3 \quad (3)$$

Given that larger individuals are often superior competitors and more resistant to environmental stressors and predation (Vermeij and Sandin 2008; Werner and Gilliam 1984), we assumed that both density-independent and density-dependent mortality rates decrease exponentially with host size (eqns. S6-S7). We also assumed that host growth declined with host size according to a von Bertalanffy growth curve (von Bertalanffy 1957) and that host fecundity increased with host size (eqns. S1-S5, eqn. S9). In the supplement, we describe these demographic functions in detail as well as how we translated them into estimates of S_{Hi} , G_i , and F_i for each stage.

2.2. Partner effects on host

We modified the host equations described above to account for the effects of the mutualistic partner. We considered two types of mutualisms: 1) a nutritional mutualism in which the partner increases host growth through nutrient provisioning (e.g., damselfish who fertilize host corals with their nitrogenous waste; Holbrook et al. 2008), and 2) a defensive mutualism in which the partner reduces host mortality by protecting the host from environmental stressors and/or predators (e.g., guard crabs that defend host corals against corallivores; Pratchett 2001).

For the nutritional mutualism, we made the growth terms in the host transition matrix (the G_i 's in eqn (1)) functions of partner density. These growth terms are derived from a von Bertalanffy growth equation, which describes the host diameter (D) as a function of a growth rate g , time t , and maximum diameter D_∞ :

$$D(t) = D_\infty(1 - \exp(-g(t - t_0))) \quad (4)$$

We assumed the von Bertalanffy growth parameter g is a function of the number of partners per host, P_i/H_i ; note this assumes that partners living on hosts in stage i , P_i , are evenly distributed across the host individuals. As P_i/H_i increases, the growth rate of hosts in the i^{th} stage

increases from g_0 to a maximum rate g_{max} at a rate b_g , where b_g indicates the strength of the partner's effect on host growth (eqn. (5), figure S5).

$$g\left(\frac{P_i}{H_i}\right) = g_0 + (g_{max} - g_0)\left(1 - \exp\left(-b_g\frac{P_i}{H_i}\right)\right) \quad (5)$$

For the defensive mutualism, we assumed that partners increase host survival by decreasing the hosts' density-independent mortality rate n_H (eqn. (2)). We assumed that n_H declines exponentially with increasing host area A from a maximum of $a_n + c_n$ to c_n at a rate b_n :

$$n_H(A) = a_n \exp(-b_n A) + c_n \quad (6)$$

where A represents the area of an individual host. Thus, larger hosts have a lower density-independent mortality rate (figure S1). The host's asymptotic density-independent mortality rate c_n is a function of P_i/H_i : as P_i/H_i increases, c_n decreases from a baseline c_{n0} to a minimum value $c_{n,min}$ at a rate b_{cn} , where b_{cn} determines the strength of the partner's effect on host density-independent mortality (eqn. (7), figure S6).

$$c_n\left(\frac{P_i}{H_i}\right) = (c_{n0} - c_{n,min})\exp\left(-b_{cn}\frac{P_i}{H_i}\right) + c_{n,min} \quad (7)$$

2.3. Partner dynamics

In our model, the partner population is structured according to the host stages that the partners are living on; for simplicity we ignore stage structure among partner individuals. We assumed that the primary benefit of associating with the host is reduced density-dependent mortality and that the strength of this benefit increases with host size (e.g., larger hosts provide greater structural refuge and thus reduce competition for shelter from predators; Holbrook and Schmitt 2002). Given that partners often rely on the three-dimensional structure provided by their host (Holbrook and Schmitt 2002), we assumed a linear relationship between the volume of an individual host and the maximum number of partners it can support, which we refer to as its carrying capacity K (Detmer et al. 2022). The partner's per capita density-dependent mortality rate m_P is proportional to $1/K$. Converting this relationship from a

Table 1

Model state variables and parameters. The parameters A_{min} , A_{rep} , and A_{max} were based on available studies for *Pocillopora* corals (Stimson 1978; Speare et al., 2022). For the remaining parameters, values were chosen that produced reasonable model dynamics, but full sensitivity analyses were performed in the supplement.

Symbol	Description	Units	Default Value
State variables			
H_i	Number of hosts in stage i	no. individuals	-
P_i	Partner density on hosts in stage i	no. individuals	-
t	Time	months	-
Parameters			
A_{patch}	Area of habitat patch	cm ²	186,532
A_{min}	Minimum host area	cm ²	0.785
A_{rep}	Area at which hosts reach reproductive maturity	cm ²	153.9
A_{max}	Maximum host area	cm ²	1,963
g_0	Min. host growth rate	month ⁻¹	0.025
g_{max}	Max. host growth rate	month ⁻¹	0.1
b_g	Rate of increase in host growth with per-host partner density	(no. P/no. H) ⁻¹	1
a_n	Max. possible decrease in host density-independent mortality rate with increasing host size	month ⁻¹	0.15
b_n	Rate of decrease in host density-independent mortality with host area	cm ⁻²	0.005
c_{n0}	Max. baseline asymptotic host density-independent mortality rate	month ⁻¹	0.05
$c_{n,min}$	Min. baseline asymptotic host density-independent mortality rate	month ⁻¹	0.001
b_{cn}	Rate of decrease in asymptotic host density-independent mortality rate with per-host partner density	(no. P/no. H) ⁻¹	1
a_m	Max. possible decrease in host density-dependent mortality rate with increasing host size	month ⁻¹	0.15
b_m	Rate of decrease in host density-dependent mortality with host area	cm ⁻²	0.005
c_m	Asymptotic host density-dependent mortality rate	month ⁻¹	0.05
F_{const}	Fecundity of a host at maturation	month ⁻¹	0.25
F_{exp}	Rate of increase in host fecundity with host area	cm ⁻²	0.00025
K_0	Theoretical partner carrying capacity on a host with no area	no. P	0.005
s_{mP}	Rate at which partner carrying capacity increases with host volume	no. P cm ⁻³	0.0015
n_P	Partner density-independent mortality rate	month ⁻¹	0.35
r_P	Partner fecundity	month ⁻¹	1
ϕ	Proportion of partner recruits that avoid juvenile hosts	-	evolved/ varied

function of host volume to a function of host area results in the following expression for partner density-dependent mortality (also see figure S7):

$$m_P(A) = (s_{mP}A^{3/2} + K_0)^{-1} \quad (8)$$

Here, A is the area of an individual host, s_{mP} determines the rate at which the partner's per capita density-dependent mortality declines with host area and K_0 is roughly equivalent to partner carrying capacity on a hypothetical host with an area of zero. We estimated the density-dependent mortality rate of partners living on hosts within stage i , $\bar{m}_{Pi}$, by averaging the function $m_P(A)$ over the range of host areas within that stage, again assuming that host sizes are uniformly distributed within a stage.

We also assumed that partners experience a constant density-independent mortality rate n_P that is independent of their host's size. Thus, the overall probability of survival of partners on hosts in stage i is given by

$$S_{Pi} = \exp \left(- \left(n_P + \bar{m}_{Pi} \frac{P_i}{H_i} \right) \right) \quad (9)$$

We assumed the death of a host also results in the death of the partners living on that host, meaning overall partner survival probability is $S_{Pi}S_{Hi}$. Partners will transition to a larger stage when their hosts grow into that stage—i.e., the partner transition probabilities are the transitions for the host multiplied by partner survival. Each partner produces r_P new recruits (larvae that survive to settle on a host) every timestep, which are assumed to all go into a common pool regardless of the host stage from which they originated. Letting R_{tot} represent the number of new partner recruits, the number of partners on each stage at time $t+1$ is given by

$$\vec{P}_{t+1} = \begin{bmatrix} (1-G_1)S_{P1}S_{H1} & 0 & 0 \\ G_1S_{P1}S_{H1} & (1-G_2)S_{P2}S_{H2} & 0 \\ 0 & G_2S_{P2}S_{H2} & S_{P3}S_{H3} \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix}_t + \begin{bmatrix} x_1R_{tot} \\ x_2R_{tot} \\ x_3R_{tot} \end{bmatrix} \quad (10)$$

where x_i represents the relative settlement onto hosts in the i^{th} stage (determined by partner behavior; see below) and R_{tot} is calculated as

$$R_{tot} = r_P \sum_{i=1} P_i \quad (11)$$

The number of recruits settling into stage i is a function of the total number of recruits and partner behavior. We described partner behavior with a single parameter ϕ , representing the fraction of recruits that ignore juvenile hosts. When $\phi = 0$, recruits distribute themselves across host stages proportional to the area covered by each stage. When $\phi = 1$, recruits distribute themselves across only adult (reproductively mature) stages (H_2 and H_3) proportional to the area covered by each adult stage. Thus, the proportion of total recruits settling on juvenile hosts (H_1) is given by

$$x_1 = (1-\phi) \frac{H_1 A_{mid,1}}{A_{tot}} \quad (12)$$

where H_1 is the number of individuals in host stage 1 at time t , $A_{mid,1}$ is the area of an average host individual in stage 1 (assumed to be the midpoint of the range of areas in this stage) and A_{tot} is the total area covered by hosts across stages at time t . The proportion of total recruits settling on hosts in the adult stages j ($j = 2$ or 3) is given by

$$x_j = (1-\phi) \frac{H_j A_{mid,j}}{A_{tot}} + \phi \frac{H_j A_{mid,j}}{H_2 A_{mid,2} + H_3 A_{mid,3}}, \quad j \in (2, 3) \quad (13)$$

We assumed the survival of newly settled recruits on each host stage declined with the density of existing partners on this stage (eqn. (9)).

3. Analyses and results

All model simulations were performed in R (v4.3.1; R Core Team 2023). In all analyses, we assumed the model had equilibrated if the change in the partner and host populations over the last 10 time steps were both $<0.1\%$. We also checked for multiple equilibria (initial condition dependence) and found none (e.g., see figures S9 and S13).

3.1. Partner and host equilibria vary with mutualism strength and the partner's behavioral strategy

We began exploring our model's ecological dynamics in the absence of any partner stage preferences (i.e., $\phi = 0$). For both the nutritional and defensive mutualisms, partner and host populations reached stable equilibria that increased with the strength of the partner's effects on its host (figure S8). When partners benefitted the hosts, equilibrium abundances of the partner and host were higher for the defensive mutualism than the nutritional mutualism. In defensive mutualisms, partners increased host survival and thereby indirectly increased their

own survival as well (figure S8). Although increasing host growth indirectly facilitates host and partner survival because host- and therefore partner- mortality decline with host size, this effect of nutritional mutualisms is smaller than the direct effect of the defensive mutualist.

Our main question of interest was the degree to which mutualistic partners should prefer to associate with (higher quality) adult hosts. We first addressed this question by investigating how equilibrium partner density and host cover varied as a function of the strength of partner preference for adult hosts (ϕ) and the strength of the partner's effects on the host. Partner effects were determined by b_g for nutritional mutualisms and b_{cn} for defensive mutualisms; these control, respectively, the rates at which host growth and survivorship increase with per-host partner density.

Both partner and host equilibria were sensitive to ϕ (Fig. 2), particularly for nutritional mutualisms. Across all mutualism strengths considered, partner density varied unimodally with ϕ and was generally maximized at lower ϕ values for nutritional mutualisms than for defensive mutualisms. For example, at our default mutualism strengths ($b_x = 1$), partner density was highest at $\phi \approx 0.86$ for the defensive mutualism and $\phi \approx 0.17$ for the nutritional mutualism.

3.2. Evolution simulations suggest evolutionarily stable partner behaviors do not always maximize partner abundance

The previous simulations only considered ecological dynamics. The resulting ϕ values represent behaviors that balance the benefits of concentrating on highest quality hosts with the benefits of spreading out (reducing crowding and distributing partner benefits across hosts) in a

way that maximizes partner population size. To determine whether these partner-maximizing trait values were also evolutionarily stable, we next allowed ϕ to evolve within the partner population using an adaptive dynamics approach. Evolutionary trajectories were modeled as a series of replacement events that occurred when mutants (individual partners possessing a trait value- in this case ϕ - that differed slightly from that of the resident population) invaded the system and potentially replaced the resident (Brännström et al. 2013). We began with a resident partner population with $\phi = 0$ (no preference for adult hosts) and allowed the resident partner and host populations to equilibrate. We considered the model to have equilibrated if both populations changed by $<0.1\%$ over the last 10 time steps of the simulation. We then introduced a small number of mutant partners with $\phi = 0.005$ and calculated their growth rate (change in number of mutant partners over time) when rare. If this growth rate was positive, we assumed that the mutant would successfully invade and eventually become the new resident (Brännström et al. 2013). We then let the model equilibrate- i. e., run until the changes in host and partner population sizes over the final 10 time steps were both $<0.1\%$ - with the new resident's ϕ value (ϕ_R) and repeated the process, but this time using mutants with ϕ values of either $\phi_R + 0.005$ or $\phi_R - 0.005$. This continued until a resident's ϕ value could not be invaded, which we assumed to be the evolutionarily stable value (figure S10).

In our adaptive dynamic simulations, the partner evolved a strong preference for adult hosts regardless of mutualism type ($\phi \approx 0.9$; Fig. 3, figure S10), and these values were globally stable (i.e., resistant to invasion by mutants with any other ϕ value; figure S11). The ESS decreased to approximately 0.8 when partner density dependent mortality was increased (figure S12), but in both cases the ESS did not vary

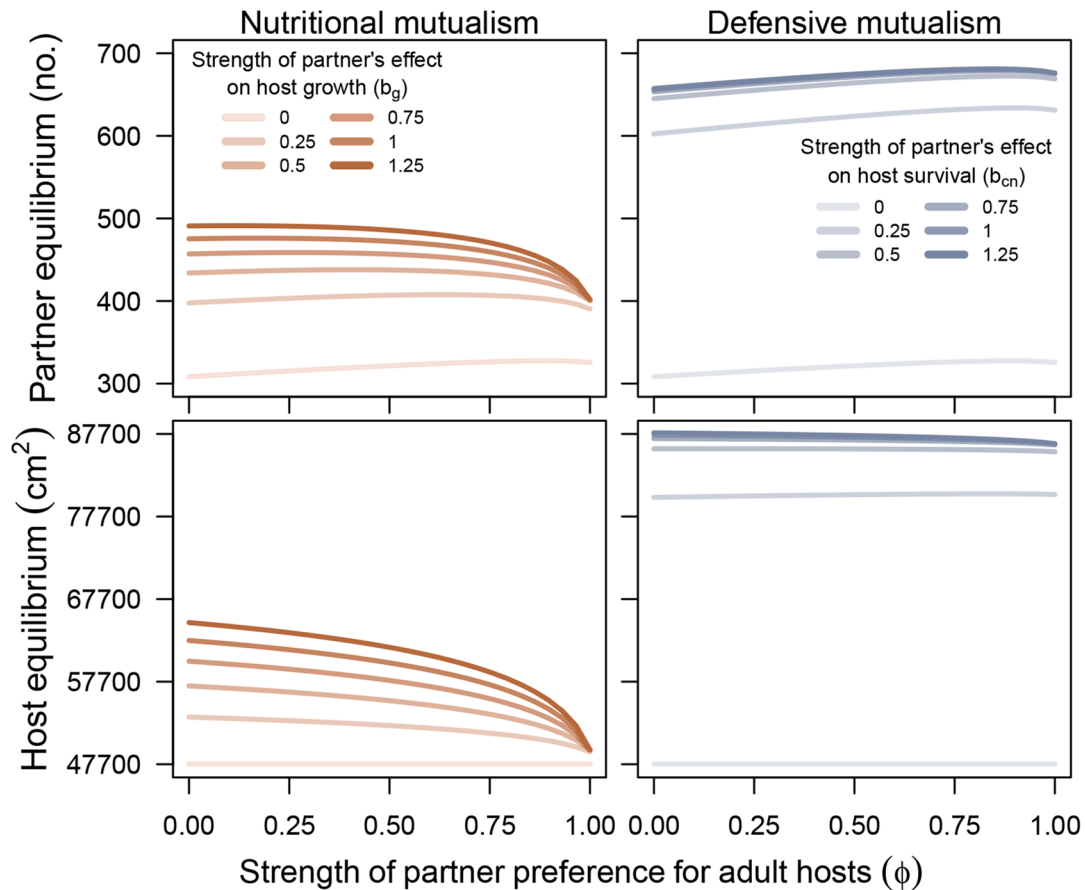


Fig. 2. Total partner density (top row) and total host cover (bottom row) as a function of the strength of the partner's preference for adult hosts ϕ . Left column: nutritional mutualism (b_g varies from 0 to 1.25, $b_{cn} = 0$); right column: defensive mutualism ($b_g = 0$, b_{cn} varies from 0 to 1.25). Colors in this figure and subsequent figures are from the Manu: NZ Bird Colour Palettes by Thomson 2022.

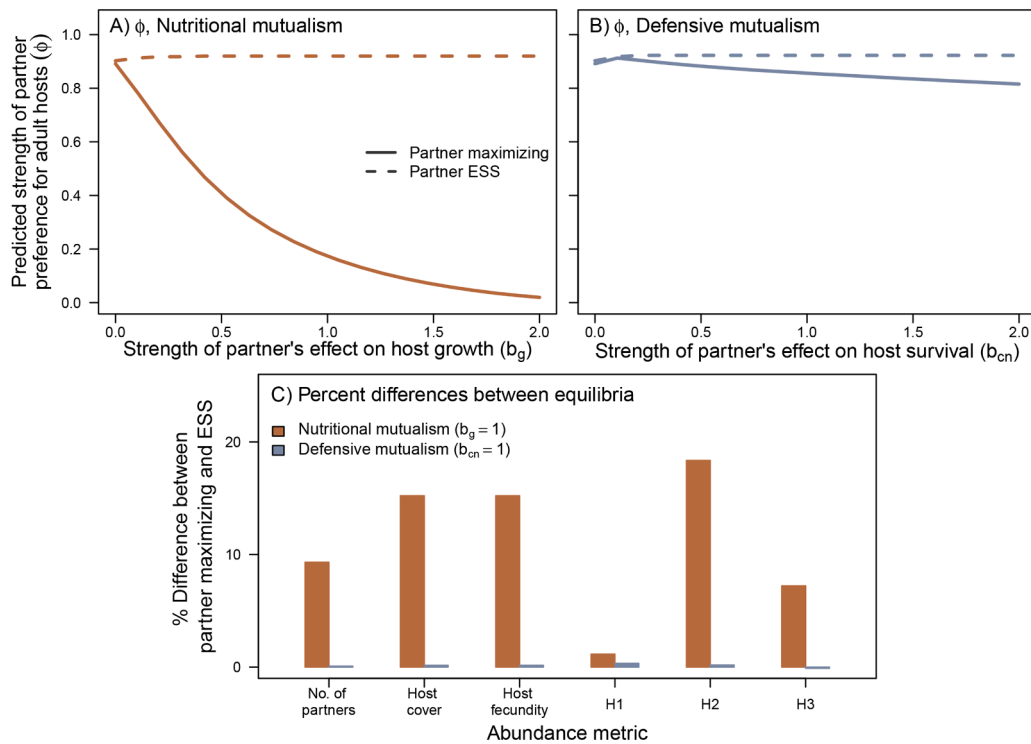


Fig. 3. A-B) Partner maximizing and evolutionary stable (ESS) ϕ values for nutritional ($b_g > 0$, $b_{cn} = 0$) and defensive ($b_g = 0$, $b_{cn} > 0$) mutualisms of varying strengths. In A) the x-axis is the rate at which host growth rate increases with partner density, while in B) it is the rate at which host density-independent mortality rate decreases with partner density. Solid and dashed lines represent ϕ values resulting in the highest possible equilibrium partner density and evolutionarily stable behaviors, respectively (see figure S12 for host-maximizing ϕ values). C) Comparison of model equilibria at partner maximizing and ESS ϕ values. The y-axis is calculated as the difference between abundances at the partner maximizing and ESS ϕ values divided by abundance at the ESS ϕ value. Positive values indicate a metric was higher at the partner maximizing ϕ value than the ESS value. Abundance metrics include partner population size, total area covered by hosts, total host reproductive output, and number of hosts in each stage (to show differences in host size structure).

greatly with the strength of the partner's effects on its host (Fig. 3, figure S12). To better evaluate the population-level consequences of these ESS ϕ values, we compared population equilibria under ESS and those attained at the ϕ value that maximized partner population size. For the defensive mutualism, the partner maximizing ϕ tended to be similar to the ESS, although it declined to slightly below the ESS when the mutualism was strong (Fig. 3B). In contrast, for the nutritional mutualism, the partner maximizing ϕ tended to be much lower than the ESS, especially when the partner had a strong effect on host growth (Fig. 3A). This difference between partner-maximizing and evolutionarily stable ϕ values corresponded to relatively large differences in partner and host equilibria, with the ESS leading to lower partner density and host cover than the partner maximizing ϕ value (Fig. 3C, figure S13).

Although our main interest was the partner behaviors that are best for the partner, we also calculated the value of ϕ that maximized host cover to compare partner behaviors that were best for the partner to those that were best for the host. For some mutualism strengths, the host maximizing ϕ value was very different from the partner maximizing and partner ESS ϕ values. For example, in the nutritional mutualism, the host maximizing ϕ value was always zero, indicating that hosts benefited most when partners distributed themselves across all host stages (figure S12). However, across all mutualism types and strengths, equilibrium partner density was generally quite similar when ϕ was at either the partner or host maximizing values, and this was true for host cover as well (figure S14, S15). We therefore focused on the partner maximizing and partner ESS values in the main text.

Sensitivity analyses indicate that the ESS ϕ values were robust to local variation in model parameters, as was the decrease in host and partner equilibria in nutritional mutualisms when ϕ was at ESS values compared to partner-maximizing values (figure S16-S18). These results also held when the number of host stages was increased from three to ten

(figure S19). In all cases, the difference between the partner maximizing and ESS ϕ values for the defensive mutualism was negligible, as were the partner and host equilibria at these ϕ values.

3.3. Model predictions are consistent with empirical data

The model analyzed above describes a theoretical mutualistic system in which host quality increases with host size. As an initial step towards testing this model's predictions, we explored empirical data on branching corals (hosts) and associated damselfish (partners). We used unpublished survey data associated with Holbrook et al. (2008) that were obtained from scuba surveys conducted by Sally J. Holbrook and Russell J. Schmitt in 2005 in mid-lagoon areas between Opunohu Bay and Irihoru Pass on the north shore of Moorea, French Polynesia (17° 32' S, 149° 50' W). These data include the length and width of individual hosts (colonies of coral in the genus *Pocillopora*) and the number of partners (three species of damselfish) on each colony. We converted colony sizes to their footprint by assuming an elliptical planar area; i.e., $\text{area} = \pi \times \text{length} \times \text{width} / 4$. Because we were interested in whether damselfish display a preference for larger colonies, we compared the size distribution of occupied colonies (those with at least one damselfish) to the size distribution of all colonies. We also investigated how the number of damselfish per colony varied with colony area and whether the colony was a juvenile or adult, assuming *Pocillopora* mature at a diameter of 14 cm = area of $\sim 150 \text{ cm}^2$ (Stimson 1978).

We found that a disproportionately low number of small coral colonies were occupied by damselfish; furthermore, almost no damselfish individuals were observed on juvenile colonies (Fig. 4). This pattern was not due to the smaller area occupied by small corals, as preferences (fish number relative to the area occupied by each coral size-class) were essentially zero for juvenile corals and increased for adult corals (figure

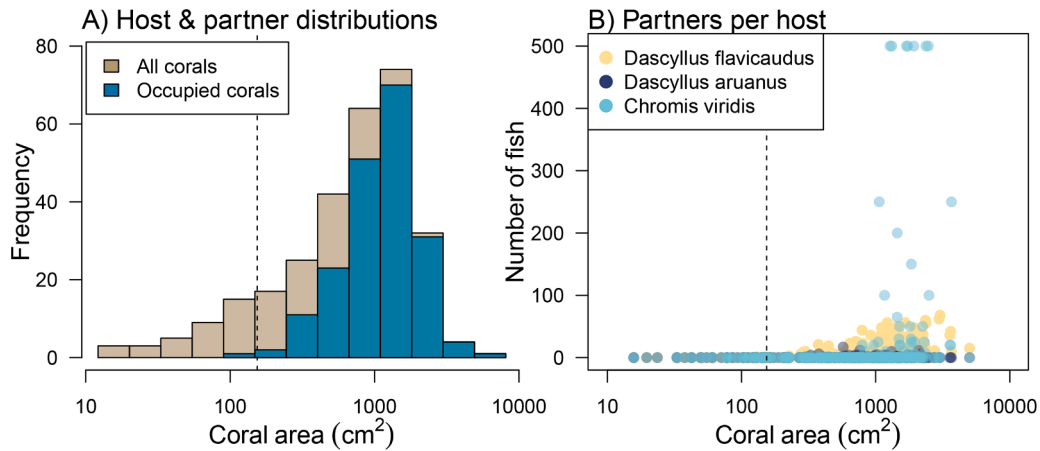


Fig. 4. Empirical data from a survey of coral-associated fishes on *Pocillopora* corals in Moorea, French Polynesia (data collected by Sally Holbrook and Russel Schmitt). A) Size distribution of corals surveyed (brown) and the subset of these corals hosting at least one damselfish (blue). B) Area of each coral in the survey (x-axis) and number of individuals of three species of damselfish observed on each coral (y-axis). Coral area is plotted on a log scale to highlight the lack of damselfish on the smallest corals. The dashed vertical line indicates the size at which *Pocillopora* corals are thought to become reproductively mature in this system (Stimson 1978).

S20). While a trend towards increasing fish abundance on larger corals could be achieved with post-settlement processes alone (e.g., higher mortality on juvenile corals), inclusion of evolutionarily stable partner behaviors in our model improved congruence with the empirical data (figure S21).

3.4. Mismatch between partner maximizing and evolutionarily stable behavioral strategies has consequences for recovery from disturbance

Our model predicts that partners should exhibit a strong preference for adult hosts (as this is their evolutionarily stable strategy), and this prediction is consistent with empirical data on branching corals and their damselfish partners. However, for nutritional mutualisms, the model also predicts ESS behavior will lead to partner densities that are

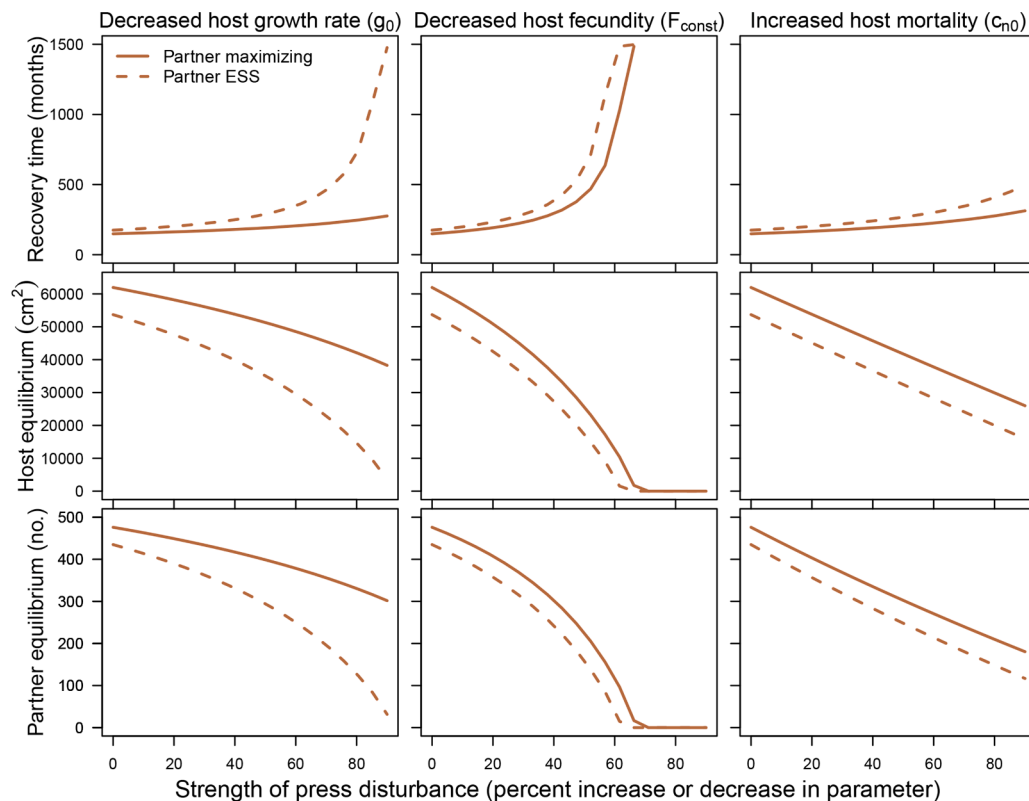


Fig. 5. Recovery times and post-disturbance equilibria at partner maximizing (solid lines) and evolutionarily stable (dashed lines) ϕ values. All results are for a nutritional mutualism ($b_g = 1$, $b_{cn} = 0$). The x-axis represents the strength of the press disturbance, indicated by the percent decrease in baseline host growth rate g_0 (left column), percent decrease in host fecundity rate F_{const} (middle column), or percent increase in asymptotic host density-independent mortality rate c_{n0} (right column). The top row shows the number of months it took the system to recover from a severe pulse disturbance, while the middle and bottom rows show total host cover and number of partners, respectively, at the post-disturbance equilibrium. Recovery times are not shown if the host and partner populations declined to zero.

appreciably lower than would be achieved with a weaker preference for adult hosts; i.e., the partner maximizing ϕ is less than the ESS ϕ . We next explored whether ESS behavior could also have consequences for recovery from disturbance and responses to environmental degradation. To do this, we ran simulations in which both host and partner populations were disturbed (initialized at low abundances) and recorded the time it took for the system to recover. In addition to these pulse disturbances, we also explored the impacts of chronic ongoing environmental degradation (a press disturbance) by simulating varying levels of either 1) decreased baseline host growth rate, 2) decreased host fecundity, or 3) increased host density-independent mortality rate. These simulations were conducted with ϕ set to either the partner maximizing or ESS values from the pre-disturbance environment, but the same initial conditions were used in both scenarios. We used initial conditions corresponding to a 99 % loss of hosts in each size class and their associated partners relative to the pre-disturbance ESS equilibria. We defined recovery as the first time point when total population sizes (i.e., the sum across all host stages) of hosts and partners were both within 1 % of their post-disturbance equilibria. This definition means the population sizes of recovered populations varied with the post-disturbance equilibria, and a population with a higher post-disturbance equilibria would therefore require a larger increase in population size to be considered “recovered”.

As expected, higher levels of post-disturbance degradation (i.e., lower host growth rate, lower host fecundity, or higher host mortality rate) resulted in longer recovery times and lower post-disturbance equilibrium host cover and partner density (Fig. 5, figure S22, figure S23). For nutritional mutualisms, recovery times were longer and equilibrium were abundances lower when ϕ was at the ESS value than when it was at the partner maximizing value; this difference was particularly pronounced when the press disturbance caused strong reductions in baseline host growth rates (Fig. 5). Furthermore, across all forms of environmental degradation, recovery times increased more steeply with worsening environmental conditions when ϕ was at the ESS value (Fig. 5). Thus, the ESS ϕ not only resulted in lower post-disturbance equilibria than the partner maximizing ϕ , but also caused populations to take longer to reach these post-disturbance equilibria—indicating that ESS behaviors resulted in reduced growth rates relative to partner maximizing behaviors. In contrast, for defensive mutualisms recovery times and post-disturbance equilibria were essentially identical for both ESS and partner maximizing behaviors (figure S23).

4. Discussion

Choice of hosts by partners is a fundamental component of mutualistic interactions, but much of our current understanding of partner choice ignores intraspecific ontogenetic variation among host individuals (but see Nakazawa et al. 2023). Here, we used a model of a stage-structured host and its mutualistic partner to explore intraspecific partner choice in a system where host quality increases throughout its ontogeny. We found that partners generally evolved to have a strong preference for the more beneficial adult hosts, but this preference could have ecological consequences in the sense that it limited population sizes and growth rates: evolutionarily stable behaviors did not always maximize equilibrium partner density or post-disturbance recovery times. These results illustrate how feedbacks between partner choice and host demographics can mediate the population- and community-level outcomes of intraspecific partner choice, providing a novel example of the important role that ontogenetic variation in host quality may play in mutualistic relationships.

A main result of this study was that partners evolved behaviors that did not always result in the maximum possible partner and host densities, indicating a conflict between the partner choices that maximize individual fitness and those that maximize population-level density. Most existing studies of eco-evo conflicts in mutualisms have focused on shifts between mutualistic and exploitative/antagonistic interactions

and the evolution of “cheaters” (Ferriere et al. 2002; Sachs and Simms 2006; Vidal et al. 2024). In these cases, conflict occurs because an individual can maximize its fitness by exploiting its partner, yet it is generally better for the population as a whole to maintain a mutually beneficial relationship (West et al. 2007). Our results highlight another source of conflict that may arise in stage-structured mutualisms: an individual partner has the highest chance of survival on high quality hosts, but distributing partner benefits across all hosts may ultimately lead to more high quality hosts and thus overall greater partner survival. Importantly, this tension arises even without any costs of the association to the host or partner, as no exploitation by either the host or partner is possible in our model.

Our model predicted that the greatest ecological costs of evolved partner preferences (i.e., greatest differences between evolved and partner-maximizing behaviors) occur in nutritional mutualisms. In these scenarios, partners that chose juvenile hosts increased the growth rate of these juveniles, shifting the host population stage structure towards more adult hosts with lower mortality rates and thereby indirectly facilitating host survival. Thus, partner population density was maximized when some partners chose (initially) lower quality hosts, even though this weaker preference for high quality hosts was not evolutionarily stable due to the higher mortality incurred by partners that recruited to small juvenile hosts of low quality. This lagged effect of partner choice on host demographics was much weaker in defensive mutualisms, where partners simply increased the survival of their chosen hosts and investing in high quality hosts was generally both evolutionarily stable and ecologically optimal. Empirical research on the role of host ontogeny in mutualistic interactions has often focused on variation in partner quality (from the perspective of the host) over a host’s lifespan and its consequences for hosts’ preferences for partners (Bronstein 1994; Yule et al. 2013; Skelton et al. 2014). For example, in ant-acacia tree mutualisms, host trees may associate with seemingly low quality ant partners because these partners become more beneficial later in the tree’s life (Palmer et al. 2010). Similarly, our model suggests that when the quality of the host (from the partner’s perspective) varies across host ontogeny, there could be population-level benefits for the partner of choosing lower quality hosts that will increase in quality as the hosts mature.

Despite both host and partner populations benefiting from partner associations with lower quality hosts, the ESS predicted by our model suggests partners should evolve a strong preference for high quality (adult) hosts. Although testing our model’s predictions will require further empirical work, we did find some supporting evidence that partners preferred high quality hosts. In the survey of *Pocillopora* corals and associated damselfish, there were disproportionately more damselfish on larger corals and essentially none on juvenile corals (Fig. 4). Damselfish can be highly selective in their choice of coral hosts (Pratchett et al. 2012), and these distributions are consistent with a strong preference for adult corals. As another example, a study on the mutualism between defensive ants and tree cholla cactus found that host plant quality (quality and quantity of extrafloral nectaries) increased with plant size, as did the probability of a plant being occupied by the mutualistic ant partner (Miller 2014). Plant-ants are known to use volatile organic compounds to select high quality host plants (Orona-Tamayo and Heil 2013; Razo-Belman et al. 2018), so these results could also indicate partner preference for larger, higher quality host individuals.

We note that while evolutionarily stable ϕ values were high in our model (generally around 0.9 across parameter values tested; Fig. 3, figure S16), they were always less than one – partners never evolved to completely ignore juvenile hosts. This result held even if the partner had no effects on the host (figure S16), indicating that it was a consequence of density-dependence in the partner population (i.e., a buffer effect; Balbontín and Ferrer 2008; Sullivan et al. 2015) rather than partner feedbacks on host demographics. Increasing the strength of partner density dependence caused evolutionarily stable ϕ values to decrease

(figure S12), further supporting the role of density dependence in weakening partner preferences for high quality hosts. Our model assumes that partners experience only negative density-dependence, e.g., due to competition for host resources (Holbrook and Schmitt 2002). While this may be accurate for strongly territorial species, positive density-dependence—specifically conspecific attraction—plays an important role in habitat selection for many taxa (Buxton et al. 2020). For example, some juvenile damselfish exhibit positive responses to conspecific cues and may therefore preferentially settle on corals already hosting conspecific individuals (Coppock et al. 2016). Such conspecific attraction would likely reduce the amount of “spillover” onto lower quality hosts in our model, potentially leading to even stronger preferences for high quality hosts due to the larger partner populations they support.

Our results could have important implications for the restoration of ecosystems with foundational species that host mutualistic partners. There is growing recognition that utilizing mutualistic interactions can aid ecosystem restoration (Derksen-Hooijberg et al., 2018; Silliman et al. 2024). For example, in coral reefs, it has been suggested that restoration projects should take advantage of positive feedbacks between coral cover and the aggregation of nutrient-providing fishes (Ladd et al. 2018). Our study suggests that the full benefits of this interaction may be unrealized in natural reefs: fish appear to prefer larger corals, but our model predicts both fish and coral populations would increase if some of these fish instead went to smaller, riskier coral hosts. Coral restoration projects could therefore potentially be enhanced by increasing fish densities on smaller corals, e.g., by manipulating the spatial configuration of small corals to redirect settling fish larvae to them (Stier and Osenberg 2010) or by manually transplanting post-settlement fish from large to small corals, which could potentially increase natural settlement on these corals due to conspecific attraction (Lecchini et al. 2007; Coppock et al. 2016). However, these approaches would first need to be tested with field experiments to determine whether the potential benefits of artificially increasing fish abundance on smaller corals outweigh the costs of greater fish mortality on these corals.

Our model makes a number of simplifying assumptions that should be considered when interpreting its predictions. Due to our focus on ontogenetic variation in host quality, we ignored size structure in the partner population. However, in some scenarios partner size and/or life stage may also play an important role in the mutualistic interaction (Nakazawa 2020). For example, in the fish-coral system, small coral hosts may offer suitable shelter for smaller, juvenile fish but not adult fish, potentially creating a “lock-and-key” type of interaction between different sized corals and fish (Brouat et al. 2001; Noonan et al. 2012). We also assumed that the host did not evolve. Incorporating host evolution and partner-host coevolution would be an important next step, particularly for mutualisms in which hosts can also exert control over the maintenance of partner-host interactions (Bronstein 2015; Pellmyr and Huth 1994; Yu et al. 2004; Edwards et al. 2006). For instance, fishes can use chemical cues from corals to find hosts (Lecchini et al. 2005; O’Connell et al. 2022) and the co-evolution of signaling between corals and their mutualistic fishes has been suggested (Stier and Osenberg 2010), although we know of no empirical evaluation of this possibility. We also only considered a single host and single partner species, but both intraspecific and interspecific partner choice (and competition among partners for hosts) may interact to shape mutualistic networks in more diverse communities (Nakazawa 2020). Future studies should explore whether the eco-evo conflicts predicted in the present study can still arise in these more realistic contexts. Finally, our model was based on coral-fish mutualisms and would need to be adapted to test the generality of our findings in other stage structured mutualisms. For example, coral-dwelling fishes typically have a pelagic larval stage that can be transported large distances by ocean currents. Partner dispersal could be very different for terrestrial species, and models of these systems might therefore need to consider spatial limitations on a partner’s ability to choose a host.

In conclusion, our theoretical model suggests that in mutualisms with ontogenetic variation in host quality, partners generally evolve strong preferences for high quality hosts; however, these behaviors may ultimately limit partner and host populations due to the lack of partner investment low quality hosts. In nutritional mutualisms, evolutionarily stable behaviors that maximized individual partner fitness increased partner and host sensitivity to disturbance relative to behaviors that maximized partner population density, highlighting the importance of considering both the evolutionary dynamics and ecological consequences of partner choice in these types of stage-structured mutualisms. Future theoretical work should explore whether these results hold when some of our model’s simplifying assumptions are relaxed; additionally, more empirical work is needed to test for partner preferences in real-world mutualisms with stage-structured hosts. Such research should improve our understanding of the roles of host ontogeny and intraspecific partner choice in mutualistic interactions.

CRediT authorship contribution statement

A.Raine Detmer: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Craig W. Osenberg:** Writing – review & editing, Methodology, Conceptualization. **Adrian C. Stier:** Writing – review & editing, Methodology, Conceptualization. **Holly V. Moeller:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank Sally Holbrook and Russell Schmitt for providing data on coral and damselfish distributions, Cherie Briggs for helpful feedback on earlier drafts of the manuscript, and members of the Moeller Lab for helpful feedback on figures. This study was funded by the National Science Foundation (grants OCE-1851510 and OCE-1851032 to A.C.S. and C.W.O., OCE-2146924 and EF-1921356 to H.V.M., and Graduate Research Fellowship # 2139319 to A.R.D.).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecolmodel.2026.111567](https://doi.org/10.1016/j.ecolmodel.2026.111567).

Data Availability

All code used in this study is available at: <https://doi.org/10.5281/zenodo.18776503>.

References

- Balbotín, J., Ferrer, M., 2008. Density-dependence by habitat heterogeneity: individual quality versus territory quality. *Oikos* 117, 1111–1114.
- Batstone, R.T., Carscadden, K.A., Afkhami, M.E., Frederickson, M.E., 2018. Using niche breadth theory to explain generalization in mutualisms. *Ecology* 99, 1039–1050.
- Bever, J.D., Richardson, S.C., Lawrence, B.M., Holmes, J., Watson, M., 2009. Preferential allocation to beneficial symbiont with spatial structure maintains mycorrhizal mutualism. *Ecol. Lett.* 12, 13–21.
- Brännström, Å., Johansson, J., Von Festerberg, N., 2013. The Hitchhiker’s Guide to adaptive dynamics. *Games* 4, 304–328.
- Bronstein, J.L., 1994. Conditional outcomes in mutualistic interactions. *Trends Ecol. Evol.* 9, 214–217.
- Bronstein, J.L., 2001. The costs of mutualism. *Am. Zool.* 41, 825–839.
- Bronstein, J.L., 2015. *Mutualism*. Oxford university press, Oxford (GB).

- Brouat, C., Garcia, N., Andary, C., McKey, D., 2001. Plant lock and ant key: pairwise coevolution of an exclusion filter in an ant-plant mutualism. *Proc. R. Soc. Lond. B: Biol. Sci.* 268, 2131–2141.
- Bshary, R., Grutter, A.S., 2002. Experimental evidence that partner choice is a driving force in the payoff distribution among cooperators or mutualists: the cleaner fish case. *Ecol. Lett.* 5, 130–136.
- Bull, J.J., Rice, W.R., 1991. Distinguishing mechanisms for the evolution of co-operation. *J. Theor. Biol.* 149, 63–74.
- Buxton, V.L., Enos, J.K., Sperry, J.H., Ward, M.P., 2020. A review of conspecific attraction for habitat selection across taxa. *Ecol. Evol.* 10, 12690–12699.
- Chamberlain, S.A., Bronstein, J.L., Rudgers, J.A., 2014. How context dependent are species interactions? *Ecol. Lett.* 17, 881–890.
- Chase, T.J., Pratchett, M.S., Walker, S.P.W., Hoogenboom, M.O., 2014. Small-scale environmental variation influences whether coral-dwelling fish promote or impede coral growth. *Oecologia* 176, 1009–1022.
- Coppock, A.G., Gardiner, N.M., Jones, G.P., 2016. Sniffing out the competition? Juvenile coral reef damselfishes use chemical cues to distinguish the presence of conspecific and heterospecific aggregations. *Behav. Process.* 125, 43–50.
- Detmer, A.R., Cuning, R., Pfab, F., Brown, A.L., Stier, A.C., Nisbet, R.M., Moeller, H.V., 2022. Fertilization by coral-dwelling fish promotes coral growth but can exacerbate bleaching response. *J. Theor. Biol.* 541, 111087.
- Derksen-Hooijberg, M., Angelini, C., Lamers, L.P.M., Borst, A., Smolders, A., Hoogveld, J. R.H., De Paoli, H., Van De Koppel, J., Silliman, B.R., Van Der Heide, T., 2018. Mutualistic interactions amplify saltmarsh restoration success. *J. Appl. Ecol.* 55, 405–414.
- Edwards, D.P., Hassall, M., Sutherland, W.J., Yu, D.W., 2006. Selection for protection in an ant-plant mutualism: host sanctions, host modularity, and the principal-agent game. *Proc. R. Soc. B: Biol. Sci.* 273, 595–602.
- Ferriere, R., Bronstein, J.L., Rinaldi, S., Law, R., Gauduchon, M., 2002. Cheating and the evolutionary stability of mutualisms. *Proc. R. Soc. Lond. B: Biol. Sci.* 269, 773–780.
- Frederickson, M.E., 2013. Rethinking mutualism stability: cheaters and the evolution of sanctions. *Q. Rev. Biol.* 88, 269–295.
- Hay, M.E., Parker, J.D., Burkepile, D.E., Caudill, C.C., Wilson, A.E., Hallinan, Z.P., Chequer, A.D., 2004. Mutualisms and aquatic community structure: the enemy of my enemy is my friend. *Annu. Rev. Ecol. Evol. Syst.* 35, 175–197.
- Holbrook, S.J., Brooks, A.J., Schmitt, R.J., Stewart, H.L., 2008. Effects of sheltering fish on growth of their host corals. *Mar. Biol.* 155, 521–530.
- Holbrook, S.J., Schmitt, R.J., 2002. Competition for shelter space causes density-dependent predation mortality in damselfishes. *Ecology* 83, 2855–2868.
- Janzen, D.H., 1966. Coevolution of mutualism between ants and acacias in Central America. *Evol. (N Y)* 20, 249.
- Ladd, M.C., Miller, M.W., Hunt, J.H., Sharp, W.C., Burkepile, D.E., 2018. Harnessing ecological processes to facilitate coral restoration. *Front. Ecol. Evol.* 16, 239–247.
- Leal, L.C., Koski, M.H., Irwin, R.E., Bronstein, J.L., 2025. Costs of floral larceny: a meta-analytical evaluation of nectar robbing and nectar theft on animal-pollinated plants. *Ecology* 106, e70036.
- Lecchini, D., Shima, J., Banaigs, B., Galzin, R., 2005. Larval sensory abilities and mechanisms of habitat selection of a coral reef fish during settlement. *Oecologia* 143, 326–334.
- Lecchini, D., Planes, S., Galzin, R., 2007. The influence of habitat characteristics and conspecifics on attraction and survival of coral reef fish juveniles. *J. Exp. Mar. Biol. Ecol.* 341, 85–90.
- Miller, T.E.X., 2014. Plant size and reproductive state affect the quantity and quality of rewards to animal mutualists. *J. Ecol.* 102, 496–507.
- Mueller, U.G., Poulin, J., Adams, R.M., 2004. Symbiont choice in a fungus-growing ant (Attini, Formicidae). *Behav. Ecol.* 15, 357–364.
- Muscatine, L., Porter, J.W., 1977. Reef corals: mutualistic symbioses adapted to nutrient-poor environments. *BioScience* 27, 454–460.
- Nakazawa, T., 2020. A perspective on stage-structured mutualism and its community consequences. *Oikos* 129, 297–310.
- Nakazawa, T., Katayama, N., 2020. Stage-specific parasitism by a mutualistic partner can increase the host abundance. *Front. Ecol. Evol.* 8, 602675.
- Nakazawa, T., Katayama, N., Utsumi, S., Yamawo, A., Yamamichi, M., 2023. When to help juveniles, adults, or both: analyzing the evolutionary models of stage-structured mutualism. *Front. Ecol. Evol.* 11, 1138138.
- Noonan, S., Jones, G., Pratchett, M., 2012. Coral size, health and structural complexity: effects on the ecology of a coral reef damselfish. *Mar. Ecol. Prog. Ser.* 456, 127–137.
- O'Connell, M.J., Fowler, A.M., Allan, S.J., Beretta, G.A., Booth, D.J., 2022. Recruiting a long way from home: domino damselfish *Dascyllus trimaculatus* can detect new temperate coral habitat and conspecifics. *J. Fish Biol.* 100, 1548–1552.
- Orona-Tamayo, D., Heil, M., 2013. Stabilizing mutualisms threatened by exploiters: new insights from ant-Plant research. *Biotropica* 45, 654–665.
- Palmer, T.M., Doak, D.F., Stanton, M.L., Bronstein, J.L., Kiers, E.T., Young, T.P., Goheen, J.R., et al., 2010. Synergy of multiple partners, including freeloaders, increases host fitness in a multispecies mutualism. *Proc. Natl. Acad. Sci.* 107, 17234–17239.
- Pellmyr, O., Huth, C.J., 1994. Evolutionary stability of mutualism between yuccas and yucca moths. *Nature* 372, 257–260.
- Pratchett, M., 2001. Influence of coral symbionts on feeding preferences of crown-of-thorns starfish *Acanthaster planci* in the western Pacific. *Mar. Ecol. Prog. Ser.* 214, 111–119.
- Pratchett, M.S., Coker, D.J., Jones, G.P., Munday, P.L., 2012. Specialization in habitat use by coral reef damselfishes and their susceptibility to habitat loss. *Ecol. Evol.* 2, 2168–2180.
- R Core Team, 2023. R: A language and Environment For Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
- Razo-Belman, R., Molina-Torres, J., Martínez, O., Heil, M., 2018. Plant-ants use resistance-related plant odours to assess host quality before colony founding. *J. Ecol.* 106, 379–390.
- Ricker, W.E., 1954. Stock and recruitment. *J. Fish. Res. Board Can.* 11, 559–623.
- Sachs, J.L., Mueller, U.G., Wilcox, T.P., Bull, J.J., 2004. The evolution of cooperation. *Q. Rev. Biol.* 79, 135–160.
- Sachs, J., Simms, E., 2006. Pathways to mutualism breakdown. *Trends Ecol. Evol.* 21, 585–592.
- Silliman, B.R., Hensel, M.J.S., Gibert, J.P., Daleo, P., Smith, C.S., Wiczynski, D.J., Angelini, C., et al., 2024. Harnessing ecological theory to enhance ecosystem restoration. *Curr. Biol.* 34, R418–R434.
- Simms, E.L., Taylor, D.L., 2002. Partner choice in nitrogen-fixation mutualisms of legumes and rhizobia. *Integr. Comp. Biol.* 42, 369–380.
- Skelton, J., Creed, R.P., Brown, B.L., 2014. Ontogenetic shift in host tolerance controls initiation of a cleaning symbiosis. *Oikos* 123, 677–686.
- Speare, K.E., Adam, T.C., Winslow, E.M., Lenihan, H.S., Burkepile, D.E., 2022. Size-dependent mortality of corals during marine heatwave erodes recovery capacity of a coral reef. *Glob. Chang. Biol.* 28, 1342–1358.
- Stier, A.C., Chase, T.J., Osenberg, C.W., 2025. Fish services to corals: a review of how coral-associated fishes benefit corals. *Coral. Reefs*.
- Stier, A.C., Osenberg, C.W., 2010. Propagule redirection: habitat availability reduces colonization and increases recruitment in reef fishes. *Ecology* 91, 2826–2832.
- Stimson, J.S., 1978. Mode and timing of reproduction in some common hermatypic corals of Hawaii and Enewetak. *Mar. Biol.* 48, 173–184.
- Sullivan, M.J.P., Newson, S.E., Pearce-Higgins, J.W., 2015. Evidence for the buffer effect operating in multiple species at a national scale. *Biol. Lett.* 11, 20140930.
- Thomas, M.J., Creed, R.P., Skelton, J., Brown, B.L., 2016. Ontogenetic shifts in a freshwater cleaning symbiosis: consequences for hosts and their symbionts. *Ecology* 97, 1507–1517.
- Thomson, G., 2022. Manu: NZ bird colour palettes. R package version 0.0.2.
- Vermeij, M.J.A., Sandin, S.A., 2008. Density-dependent settlement and mortality structure the earliest life phases of a coral population. *Ecology* 89, 1994–2004.
- Vidal, M.C., Agarwal, R., Segraves, K.A., 2024. Coevolution and dependency influence resistance of mutualists to exploitation. *Front. Ecol. Evol.* 12, 1113963.
- Von Bertalanffy, L., 1957. Quantitative laws in metabolism and growth. *Q. Rev. Biol.* 32, 217–231.
- Werner, E.E., Gilliam, J.F., 1984. The ontogenetic niche and species interactions in size-structured populations. *Annu. Rev. Ecol. Syst.* 15, 393–425.
- West, S.A., Griffin, A.S., Gardner, A., 2007. Evolutionary explanations for cooperation. *Curr. Biol.* 17, R661–R672.
- Younginger, B.S., Friesen, M.L., 2019. Connecting signals and benefits through partner choice in plant-microbe interactions. *FEMS Microbiol. Lett.* 366, fnz217.
- Yu, D.W., Ridley, J., Jouselin, E., Herre, E.A., Compton, S.G., Cook, J.M., Moore, J.C., et al., 2004. Oviposition strategies, host coercion and the stable exploitation of figs by wasps. *Proc. R. Soc. Lond. B: Biol. Sci.* 271, 1185–1195.
- Yule, K.M., Miller, T.E.X., Rudgers, J.A., 2013. Costs, benefits, and loss of vertically transmitted symbionts affect host population dynamics. *Oikos* 122, 1512–1520.
- Zhang, T., Jandér, K.C., Huang, J.-F., Wang, B., Zhao, J.-B., Miao, B.-G., Peng, Y.-Q., et al., 2021. The evolution of parasitism from mutualism in wasps pollinating the fig, *Ficus microcarpa*, in Yunnan Province, China. *Proc. Natl. Acad. Sci.* 118, e2021148118.